

7.2. $ax^2 + bx + c \quad a, b, c \in \mathbb{R}$

$$[\Phi]_{k_3}^{k_3} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} p'' \\ 2(ax^2)' \\ 2(bx)' \end{pmatrix} = \begin{pmatrix} 2a \\ 4a \\ 2b \end{pmatrix}$$

$$[\Phi]_{k_3}^{k_3} = \begin{pmatrix} 2 & 0 & 0 \\ 4 & 0 & 0 \\ 0 & 2 & 0 \end{pmatrix} = A$$

$$p_A(x) = (2-x)x^2$$

$\downarrow$ vl. i. 2
 alg. 1
 geo. 1

$\downarrow$ vl. i. 0
 alg. 2
 geo. 1

$$M_2 = \mathcal{LO}\left\{\begin{pmatrix} 1 \\ 2 \end{pmatrix}\right\}$$

$$M_0 = \mathcal{LO}\left\{\begin{pmatrix} 0 \\ 2 \end{pmatrix}\right\}$$

$$\vec{v}_1 = A\vec{v}_2$$

$$0 \xleftarrow{A} \vec{v}_1 \xleftarrow{A} \vec{v}_2$$

$$B = \left\{ \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \end{pmatrix} \right\}$$

$\begin{matrix} x^2+2x+2 \\ \parallel \\ 1 \\ \parallel \\ 1/2x \end{matrix}$

$$J = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} = [\Phi]_B^B$$

$$\left(\begin{array}{ccc|c} 2 & 0 & 0 & 0 \\ 4 & 0 & 0 & 0 \\ 0 & 2 & 0 & 1 \end{array} \right)$$

$$\vec{v}_2 \in \left\{ \begin{pmatrix} 0 \\ 1/2 \\ 0 \end{pmatrix} + \mathcal{LO}\left\{\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}\right\} \right\}$$

$$\vec{v}_2 = \begin{pmatrix} 0 \\ 1/2 \\ 0 \end{pmatrix}$$